KAMBHAMPATI NITHIN KUMAR

Software Developer | Web Developer | Backend Developer | Python Developer

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PROFESSIONAL SUMMARY

Results-driven Computer Science and Engineering graduate (2026) currently working as a Web Developer Intern at Magzest, where I built and deployed HRZest (www.hrzest.com), a live enterprise HR and biometric attendance platform. Hands-on experience in full stack web development, cloud deployment, machine learning, and blockchain-based cryptographic systems. Proficient in Python, Java, SQL, Flask, REST API development, Docker, and Git version control. Strong understanding of Data Structures and Algorithms, OOP, DBMS, and SDLC. Completed 4 industry internships across Web Development, Python Full Stack Development, Artificial Intelligence and Machine Learning, and Mobile Application Development. Seeking an entry-level Software Developer or Backend Developer role to apply technical skills and contribute to impactful engineering teams.

TECHNICAL SKILLS

Programming Languages: Python, Java

Web Technologies: HTML5, CSS3, JavaScript, Flask

Databases: PostgreSQL, MySQL, SQL Query Optimization

Cloud and DevOps: AWS, Docker, Nginx, Terraform, GitHub Actions (CI/CD), Linux

Version Control and Tools: Git, GitHub, REST API, JSON

Libraries and Frameworks: Scikit-learn, Pandas, NumPy, OpenCV, Flask

Core CS Concepts: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, SDLC, Agile

Specialized Knowledge: Cryptography, Digital Signatures, Blockchain Technology, Machine Learning, WebAuthn

Soft Skills: Problem Solving, Team Collaboration, Analytical Thinking, Quick Learner

PROFESSIONAL EXPERIENCE

Web Developer Intern | Magzest *Mar 2026 – Present*

- Built and deployed HRZest (www.hrzest.com), a live multi-tenant enterprise HR platform, using Python, Flask, and PostgreSQL, with 10+ modules covering employees, attendance, payroll, leave, performance reviews, onboarding, documents, support tickets, and an employee self-service portal.
- Implemented biometric attendance using face recognition (OpenCV), WebAuthn passkeys (fingerprint), QR code scanning, and GPS geo-fencing for verified employee check-in and check-out.
- Developed an automated payroll engine with salary rules, monthly reports, PDF payslips delivered by email, Excel exports, and payroll locking.
- Secured the platform with encryption of personal data at rest, bcrypt password hashing, TOTP two-factor authentication, and API rate limiting.
- Deployed to AWS using Docker containers, an Nginx reverse proxy with Let's Encrypt HTTPS, Terraform infrastructure as code, and a GitHub Actions CI/CD pipeline.

Python Full Stack Developer Intern (6 Weeks) | ExcelR Edtech Pvt. Ltd. / APSCHE *June – Aug 2024*

- Developed 3+ responsive full stack web applications using HTML5, CSS3, JavaScript, and Flask framework, following modern UI/UX and accessibility principles.
- Implemented backend logic including RESTful API design, MySQL database schema creation, query optimization, and CRUD operations handling 1,000+ records.
- Used Git and GitHub for version control, branching strategies, and collaborative development following Agile and SDLC best practices across a team of 5 interns.

Artificial Intelligence and Machine Learning Virtual Intern | SmartInternz / APSCHE *2025*

- Completed Short Term Virtual Internship Program focused on acquiring job-ready skills in Artificial Intelligence and Machine Learning through hands-on projects and assessments.
- Gained practical exposure to supervised and unsupervised learning techniques, model training pipelines, and AI application development using Python-based machine learning tools.

Mobile Application Developer Intern (Virtual) | Blackbucks Education Pvt. Ltd. *2026*

- Completed Skill Internship Program 2026 in Mobile Application Development domain, building practical skills in cross-platform mobile app design, development lifecycle, and UI implementation.
- Gained exposure to mobile application architecture, user interface design principles, and development workflows for Android and cross-platform environments.

PROJECTS

Major Project: A New Digital Signature Primitive and Its Application in Blockchain

- Designed and implemented a novel digital signature scheme with enhanced security over conventional algorithms (ECDSA, RSA) for blockchain transaction authentication, reducing signature verification overhead by approximately 20% compared to baseline RSA implementation.
- Applied advanced cryptographic primitives including elliptic curve cryptography (ECC) and zero-knowledge proof concepts to strengthen integrity and non-repudiation of digital signatures across 3 core blockchain transaction types.
- Integrated the signature mechanism into a blockchain prototype with tamper-proof, verifiable transaction signing; tested against 5+ threat models including replay attacks and signature forgery.
- Evaluated key generation, signing time, and verification speed against existing blockchain signature standards; documented findings in a formal security analysis report of 25+ pages.
- **Technologies:** Python, PyCryptodome, Elliptic Curve Cryptography (ECC), Digital Signature Algorithms, Blockchain Concepts

Minor Project: TrafficTelligence – Traffic Volume Estimation System

- Built a supervised Machine Learning model using Python, Pandas, NumPy, and Scikit-learn to estimate traffic volume achieving approximately 85% prediction accuracy across 10,000+ data records.
- Performed end-to-end data preprocessing, feature engineering on 8+ traffic variables, hyperparameter tuning, and cross-validation to reduce mean absolute error by 18%.
- Integrated the trained ML model into a Flask web application with MySQL backend; designed 4 RESTful API endpoints for dynamic data ingestion, retrieval, and real-time visualization.
- **Technologies:** Python, Scikit-learn, Pandas, NumPy, Flask, MySQL, REST API, Git, GitHub

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering *2022 – 2026*
Chebrolu Engineering College, Guntur | JNTUK | GPA: 7.0 / 10.0 | Aggregate: 70%
Relevant Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Cryptography, Machine Learning

Intermediate – Mathematics, Physics, Chemistry (MPC) *2022*
Board of Intermediate Education, Andhra Pradesh | Marks: 727 / 1000 | Percentage: 72.7% | Grade: B

Secondary School Certificate (SSC) *2020*
Board of Secondary Education, AP | Z P High School, Lakkaraju Garlapadu, Guntur | Marks: 558/600 | 93% | First Division

CERTIFICATIONS

- Python Full Stack Development – ExcelR Edtech Pvt. Ltd. in collaboration with APSCHE | Certificate No: EXCELR93061 | August 2024
- Artificial Intelligence and Machine Learning – SmartInternz Short Term Virtual Internship, APSCHE 2025
- Mobile Application Development – Blackbucks Education Pvt. Ltd. Skill Internship Program | BBID: BBSKILLINT202600196 | 2026

KEY STRENGTHS

- Strong problem-solving and analytical skills demonstrated across full stack development, a production HR platform, machine learning pipelines, and cryptographic blockchain system design.
- Proven team collaboration ability through internship work with 5-member development teams following Agile and SDLC methodologies.
- Quick learner who independently acquired skills in ECC cryptography, blockchain architecture, cloud deployment, and mobile development outside core curriculum.